

Derive: $L_{\text{edd}} = \frac{4\pi G M_{\text{BH}} m_p c}{\sigma_T} = a M_{\text{BH}}$

- AGN for our model was powered by accretion:



ϵ : radiative eff.

- $\underline{L = \epsilon \dot{M} c^2} = \frac{dE}{dt}$, $E = \epsilon M c^2$

equating: $L = \epsilon \dot{M} c^2 = \frac{4\pi G m_p c}{\sigma_T} M_{\text{BH}}$

• Find Eq. M_{BH} over time:

• Case: $L < L_{\text{edd}}$, $\dot{M} < \dot{M}_{\text{BH}}$

• $\epsilon \dot{M} = \epsilon \frac{dM}{dt}$: radiative change

• $(1-\epsilon)\dot{M} = \text{gravitational change}$

→ add

$$\epsilon \dot{M} + \underbrace{(1-\epsilon)\dot{M}} = \dot{M}$$

ϵ : radiation
→

$$\frac{dM_{\text{BH}}}{dt} = (1-\epsilon) \dot{M}$$

Solve: $\dot{M} = \frac{1}{1-\epsilon} \dot{M}_{\text{BH}}$

$$\epsilon \left(\frac{1}{1-\epsilon} \right) \dot{M}_{\text{BH}} c^2 = \frac{4\pi G m_p c}{\sigma_T} M_{\text{BH}}$$

rearrange:

$$\frac{dM_{\text{BH}}}{dt} \frac{1}{M_{\text{BH}}} = \frac{4\pi (1-\epsilon) G m_p}{\epsilon \sigma_T c}$$

$$K \equiv \frac{1}{\tau_{\text{edd}}}$$

$$\frac{dM_{\text{BH}}}{dt} \frac{1}{M_{\text{BH}}} = K$$

$$\int \frac{dM_{\text{BH}}}{M_{\text{BH}}} = \int \kappa dt$$

$$\ln(M_{\text{BH}}) = \kappa t + C \quad \Bigg| \quad M_0 = e^C$$

$$\circ \quad M_{\text{BH}}(t) = M_0 e^{\kappa t}$$

• Usually standard $\epsilon = 0.1$

$$\kappa = \frac{1}{\tau_{\text{edd}}} \quad \tau_{\text{edd}}: \text{Time scale for BH}$$

P Exam $\beta \approx (c)$:

$$M_{\text{BH}}(t_1) = M_0 e^{\kappa t_1} \quad t_1: \text{"start"}$$

$$M_{\text{BH}}(t_2) = 2 M(t_1) = M_0 e^{\kappa t_2}$$

$$2 (M_0 e^{\kappa t_1}) = M_0 e^{\kappa t_2}$$

$$2 = e^{\kappa(t_2 - t_1)}$$

$$\ln(2) = \kappa(t_2 - t_1)$$

$$\Bigg| \kappa = \frac{1}{\tau_{\text{edd}}}$$

$$\Delta t = (t_2 - t_1)$$

$$\ln(2) \tau_{\text{edd}} = \Delta t$$

• Use $\epsilon = 0.1$ $\tau_{\text{edd}} \approx 4.5 \times 10^7 \text{ yr}$

$$\ln(2) (4.5 \times 10^7 \text{ yr}) = \Delta t$$

$$\sim 3 \times 10^7 \text{ yr} = \Delta t$$

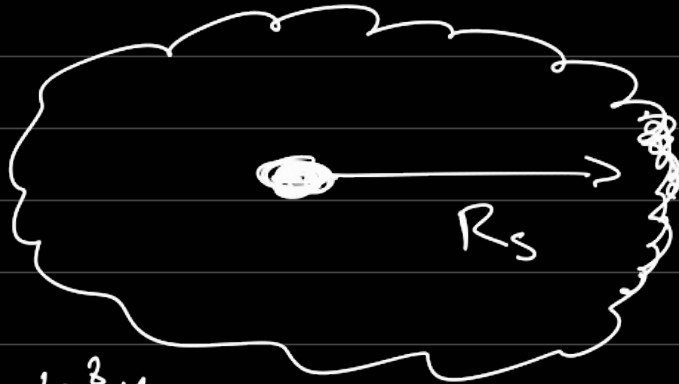
$$\tau_{\text{edd}} = \frac{1}{\kappa} = \frac{\sigma_T \epsilon c}{4\pi G m_p (1-\epsilon)} \approx 4.5 \times 10^7 \text{ yr}$$

• used $\epsilon = 0.1$

• prob. give you κ

* τ_{edd} : a time scale for a shining AGN
 $\sim 4.5 \times 10^7 \text{ yr}$

(e)



$$M_{BH} = 10^3 M_\odot$$

$$R \leq \Delta t c$$

↑
timescale for variation

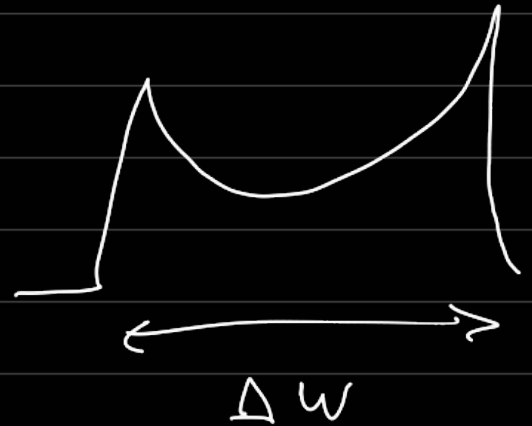
$$\frac{R_s}{c} \leq \Delta t$$

$$R_s = \frac{2 G M_{BH}}{c^2}$$

$$\frac{2 G M_{BH}}{c^3} \leq \Delta t$$

$$(c) \quad \frac{b}{a} = \cos(i)$$

$$\Delta W = 2 V_{max} \sin(i)$$



$$* V_{obs} = V_{real} \sin(i) *$$

$$(V_{obs})_1 = 600 = (V_{real})_1 \sin(i)$$

$$(V_{obs})_2 = 1000 = (V_{real})_2 \sin(i)$$

•

$$\Delta W = (V_{obs})_2 - (V_{obs})_1$$

Solve for V_{max}



V_r : V - real

V_{obs} : V - obs
 $\sin(i) = \frac{V_{obs}}{V_{real}}$

$$V_{obs} = \sin(i) V_{real}$$

PS 4) (2) $V_{obs} = 200 \text{ km/s}$

$$\frac{b}{a} = \cos(i) = 0.6$$

V_{obs} : peak vel.
 $V_{max} = V_{real} = \frac{V_{obs}}{\sin(i)}$

$$\log\left(\frac{L_v}{L_0}\right) \approx \alpha$$

• Make assumption: same $M-L$ ($\frac{M}{L}$)
 $\left(\frac{M}{L}\right) = \mu$

• Recall: in our derivation, we made lots of assumptions: one of them was on average
 $\left(\frac{M}{L}\right) = \mu$ (constant)

- spirals

$$L_M \approx M \quad \Rightarrow \quad L_v \approx \mu$$

$$\left(\frac{L_0}{M_0}\right) \approx 2$$

$$\log\left(\frac{M_v}{M_0}\right) = 11.1$$

Detail: $\frac{M_v}{L_v} = \mu = \frac{M_0}{L_0}$

$$\frac{M_v}{L_v} = \frac{M_0}{L_0} \Rightarrow \frac{M_v}{M_0} = \left(\frac{L_v}{L_0}\right)$$

$$\log\left(\frac{L_v}{L_0}\right) \approx \alpha = 3.0 \log\left(\frac{V_{max}}{200}\right) + 10.4$$

replace L

$$\log\left(\frac{M_v}{M_0}\right) = \alpha$$

$$\text{solve for } M_v = M_0 e^\alpha$$

PS4 (5) | $L(t_2) = 2L(t_1)$ ✖

$$\Delta t = t_2 - t_1 = 1 \text{ day}$$

$$L = 2 \times 10^{30} \text{ W}$$

convert to
s



✖ X-ray Source: $R \leq c \Delta t = (3 \times 10^8 \text{ m/s})(1 \text{ day})$
 $R \leq 2.6 \times 10^{13} \text{ m}$

✖ BH-Mass: Assume BH acc: assume $L = L_{\text{edd}}$

$$L = 1.26 \times 10^{31} (M_{\text{BH}}/M_0) \text{ W}$$

• solve for M_{BH}

$$\bullet \frac{M_{\text{BH}}}{M_0} = 1.6 \times 10^8 \quad \longrightarrow$$

$$R_{\text{BH}} \approx 3 R_s = 3 \left(\frac{2 G M_{\text{BH}}}{c^2} \right)$$

$$R_{\text{BH}} \approx 1.4 \times 10^{12} \text{ m}$$

$$R_{\text{BH}} < R$$

PS4 (4)

BH assume circular orbits: point source

$$v^2 = \frac{GM}{R}$$

PS4 (3) | (Monday)

PS4 optmat:

1 (b) $L = 6.3 \times 10^{13} L_{\odot}$

$$L_{\text{edd}} = 1 (M_{\text{BH}}(M_{\odot}))$$

(c) "seed" mass

$$M(t) = M_0 e^{kt}$$

$$M(t_0) = M_0 e^{kt_0}$$

$$M(t_f) = M_0 e^{kt_f} = M_{\text{BH}}$$

$$\frac{M(t_0)}{M(t_f)} = \frac{e^{kt_0}}{e^{kt_f}} = \frac{M(t_0)}{M_{\text{BH}}}$$

$$M(t_0) = M_{\text{BH}} e^{k(t_0 - t_f)}$$

$$\Delta t = t_f - t_0$$

$$M(t_0) = M_{\text{BH}} e^{-k\Delta t}$$

$$\Delta t: 0.7 \pm \text{Gyr} \quad \text{at most}$$

$$M(t_0)$$

